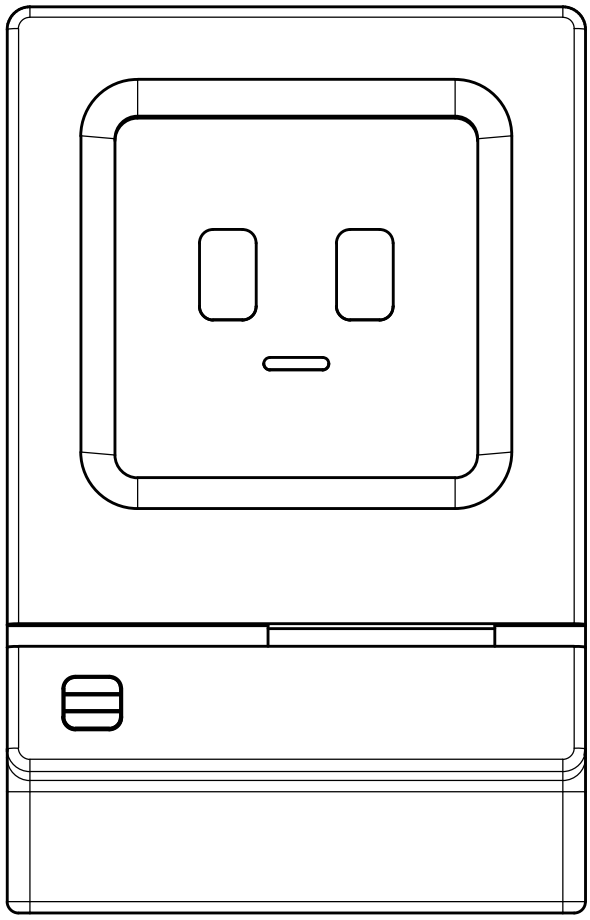
REV	DESCRIPTION	DATE
02	UPDATED FIRMWARE SETUP	JUNE 2026

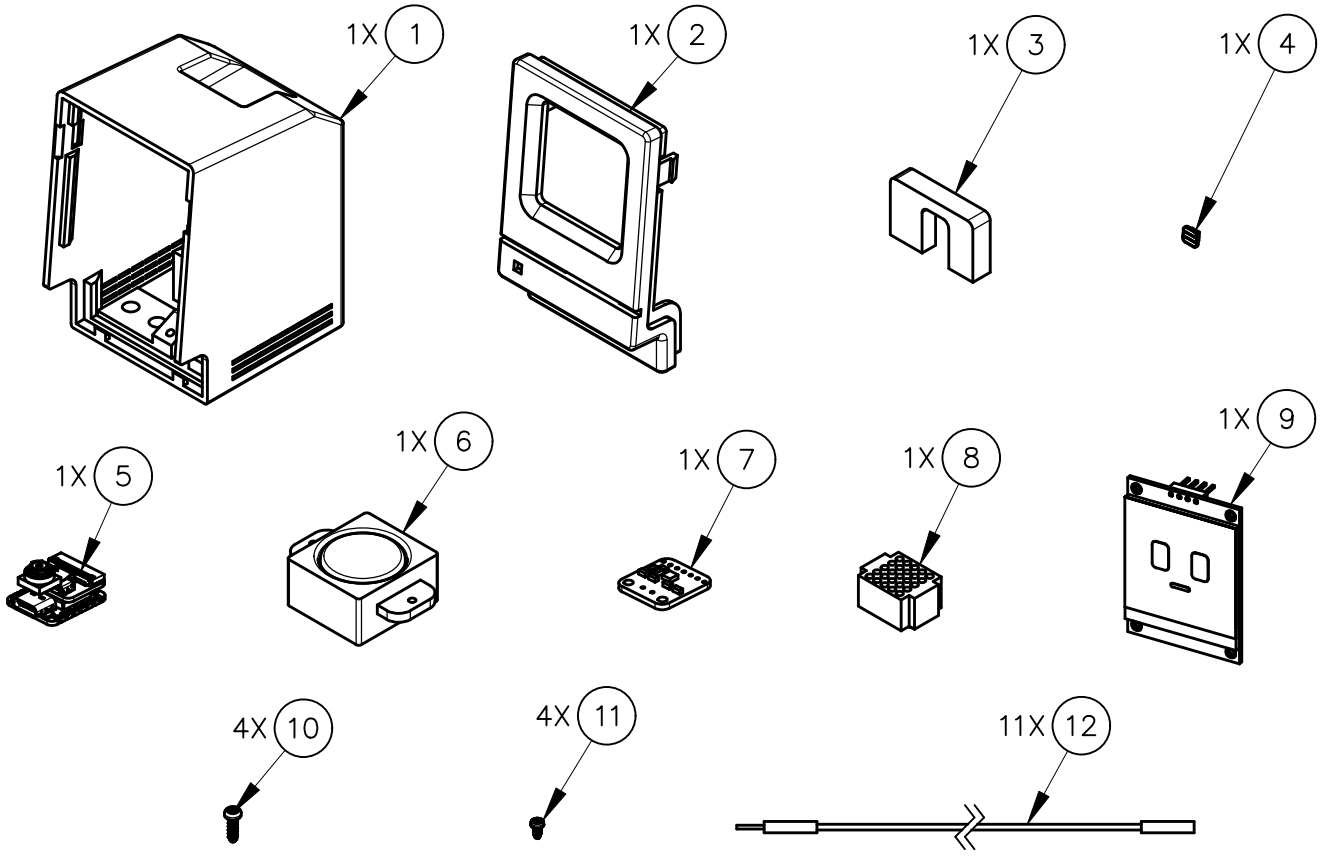


KIRA

ASSEMBLY INSTRUCTIONS

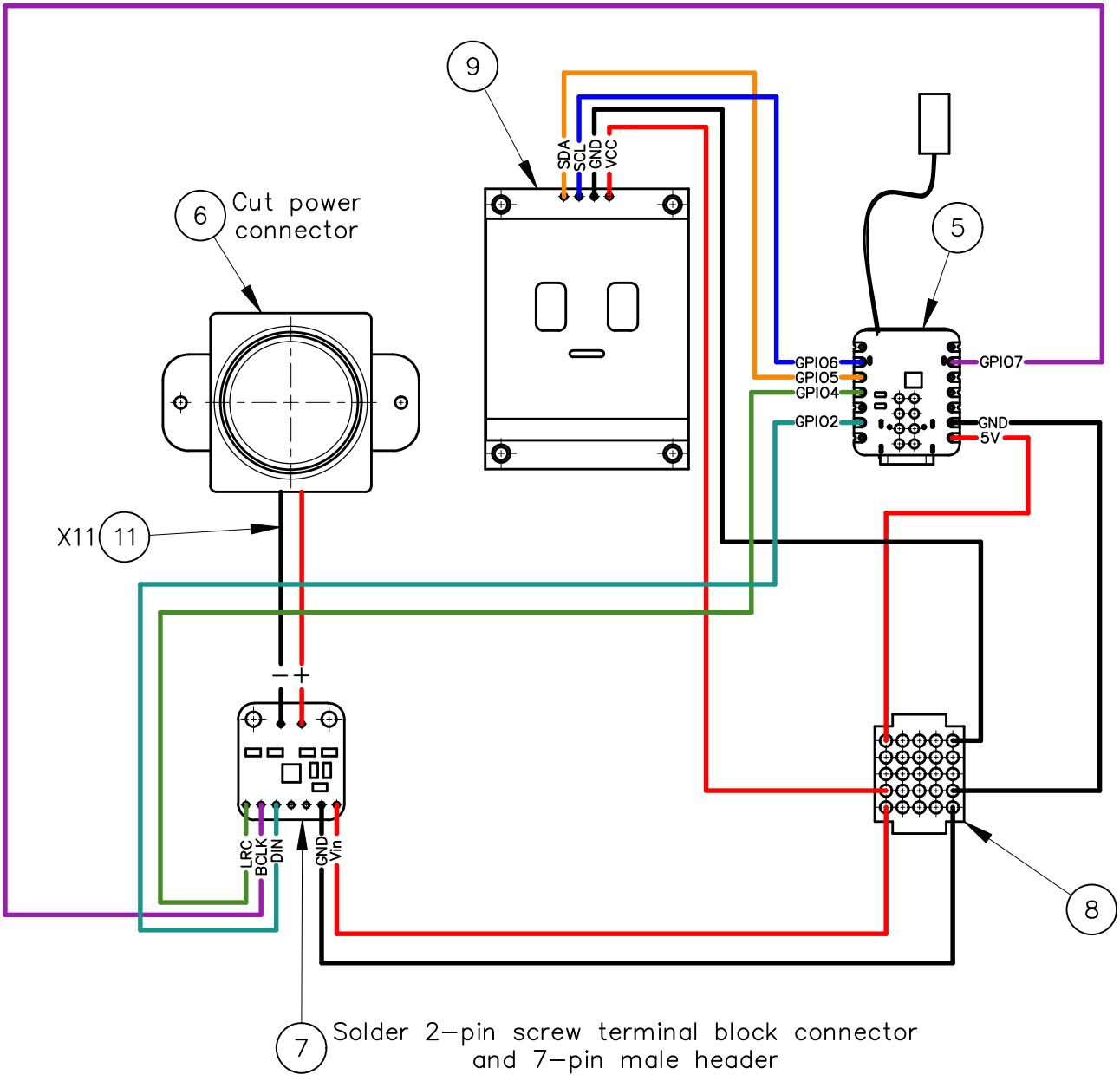
READ ALL INSTRUCTIONS
Carefuly read all instructions
before assembly and use

REVISION
02

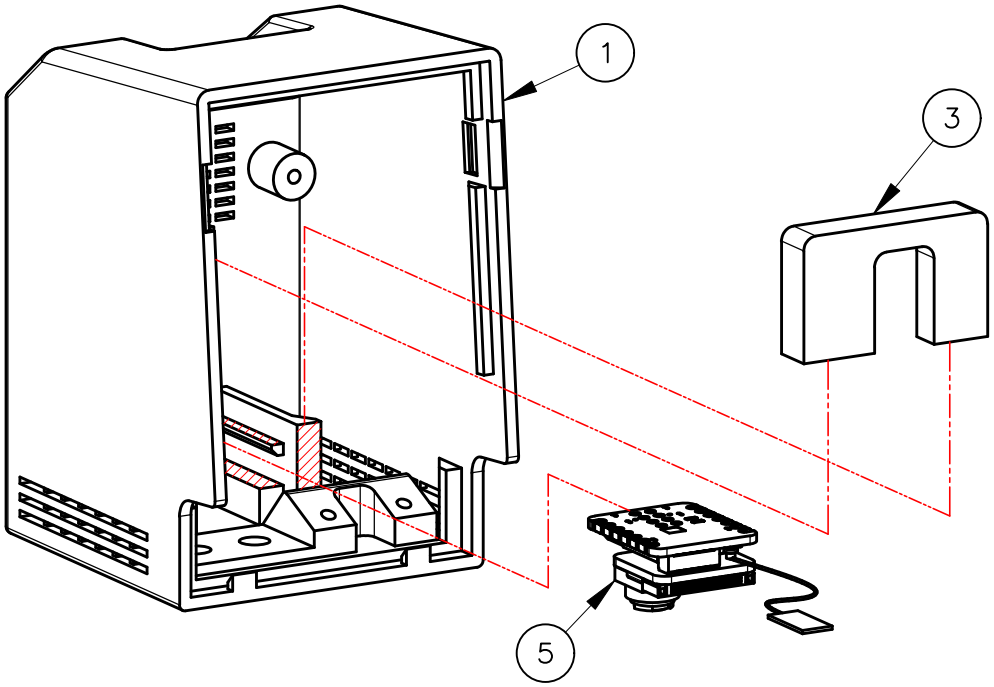


Item	Description
1	Enclosure
2	Front plate
3	Clamp
4	Badge
5	XIAO-ESP32S3-Sense v24
6	Speaker
7	Amplifier
8	Mini Breadboard
9	SH1107 OLED Display
10	M2X8mm Tapping Screw
11	M2X4mm Tapping Screw
12	Jumper

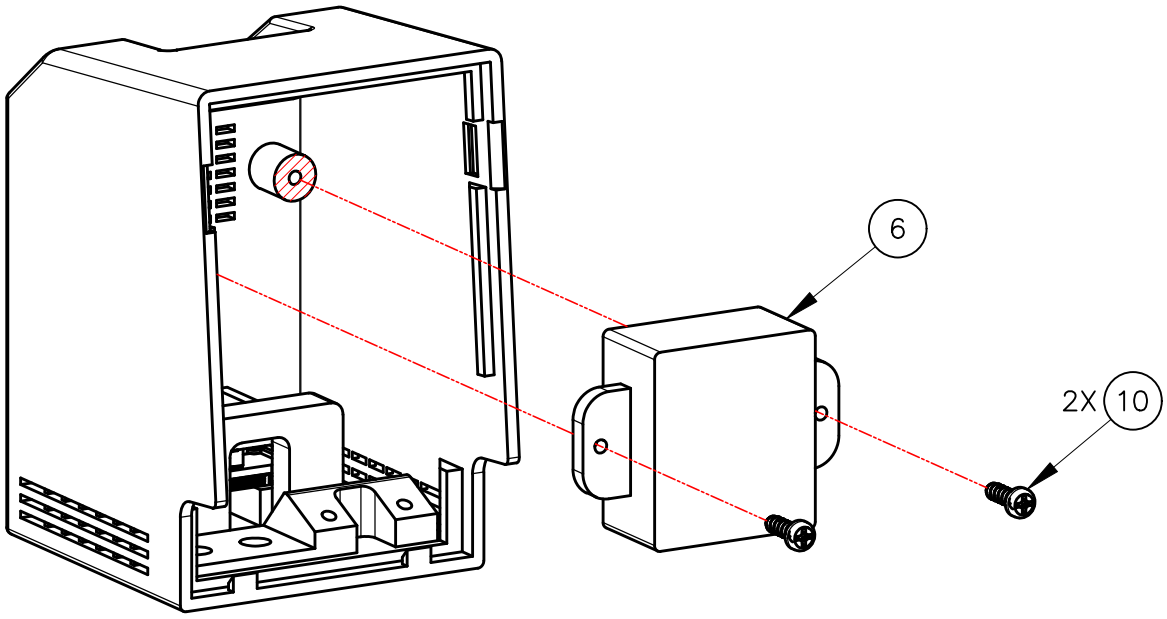
1 Use jumper wires to connect all components. After everything is connected, proceed to the next steps. Do not disconnect any components during Steps 2–8.



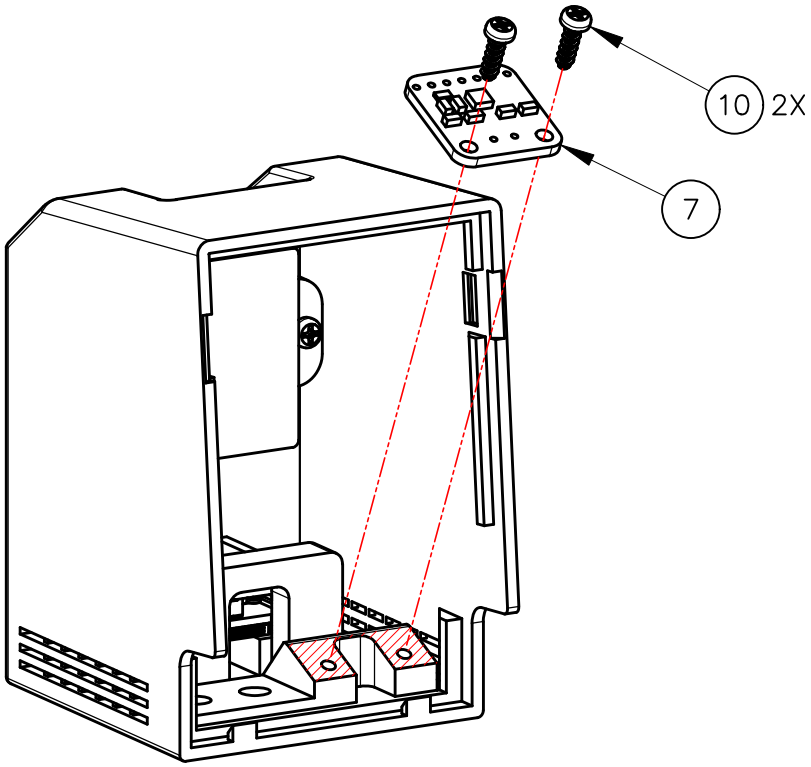
2 Mount the ESP32 on the lower rear rails of the enclosure. Route the Wi–Fi antenna through the clamp cutout, then lock the board in place with the clamp. Stick Wi–Fi antenna to any available interior surface.



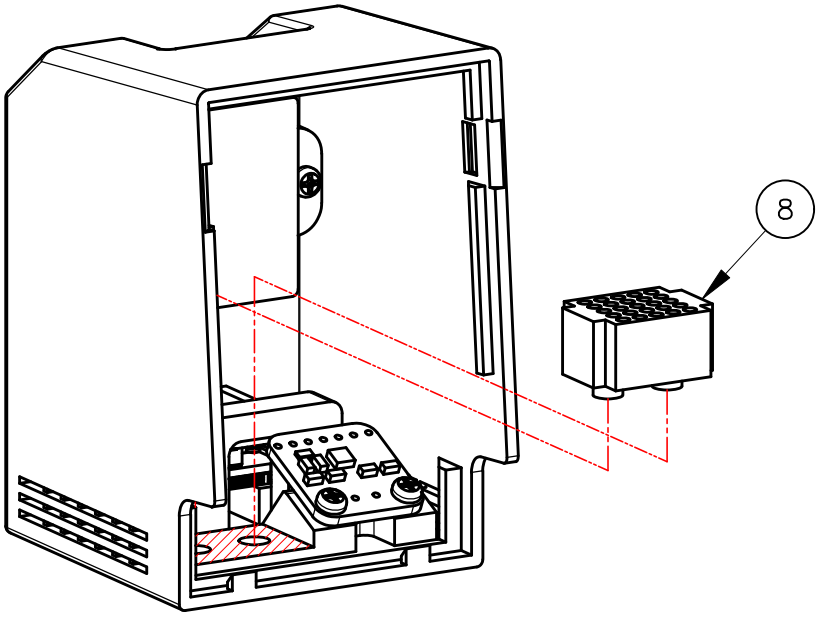
3 Mount the speaker to the two internal bosses of the enclosure using two screws.



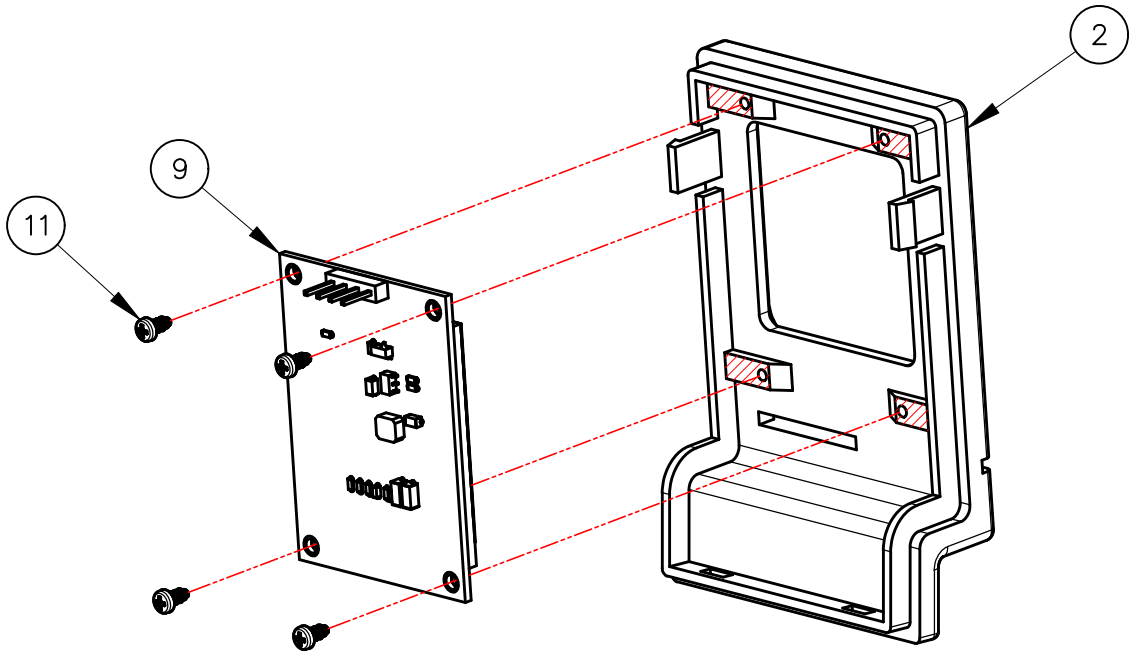
4 Mount the speaker amplifier to the angled mounting block using two screws.



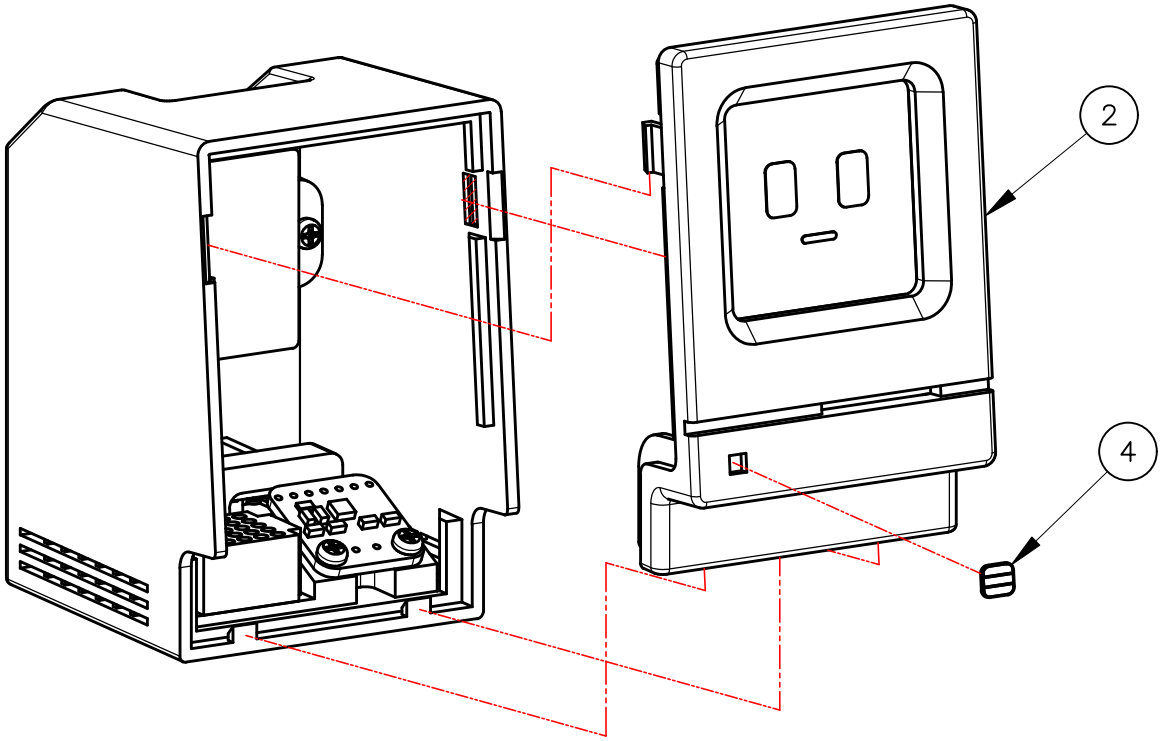
5 Press fit mini breadboard into the mounting holes. No screws required.



6 Mount the display to the front plate using four screws.



7 Align the front plate with the enclosure, insert the lower tabs, then press until the clips snap into place. Insert a badge.

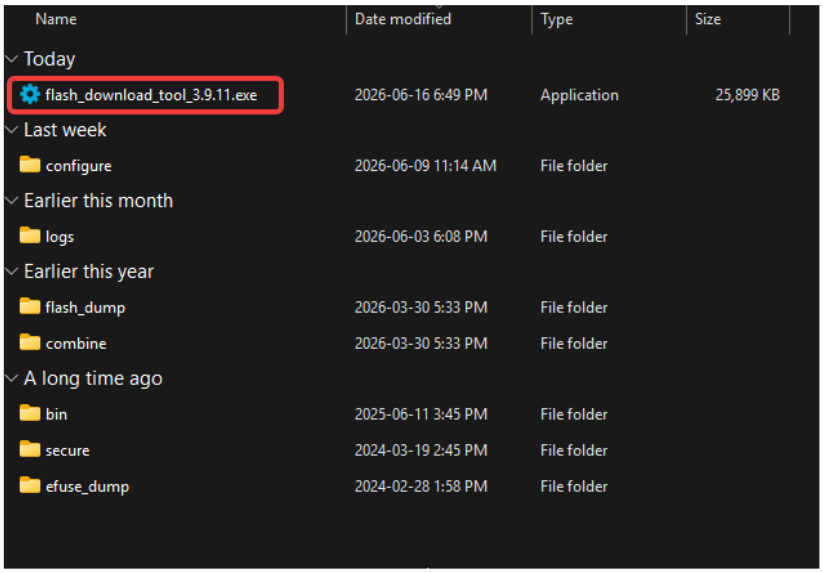


8 Setup Xiaozhi AI.

1. Download and extract Espressif flash tool (see Makerlab description for link).



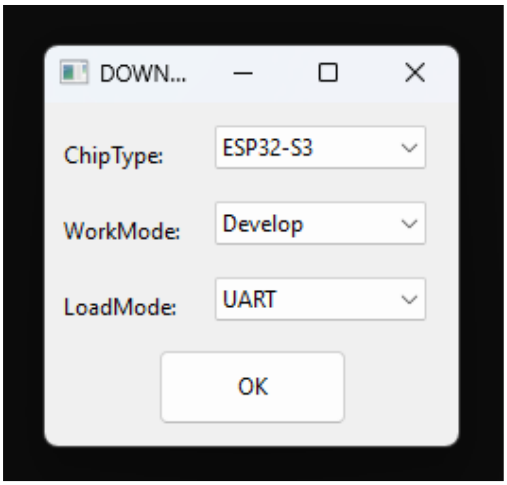
2. Download "Kira Firmware.zip" (see Makerlab description for file).
3. Extract "Kira Firmware" folder anywhere on your computer.
4. Connect Kira to your computer using a USB cable.
5. Open Espressif flash tool folder and run "flash_download_tool.exe"



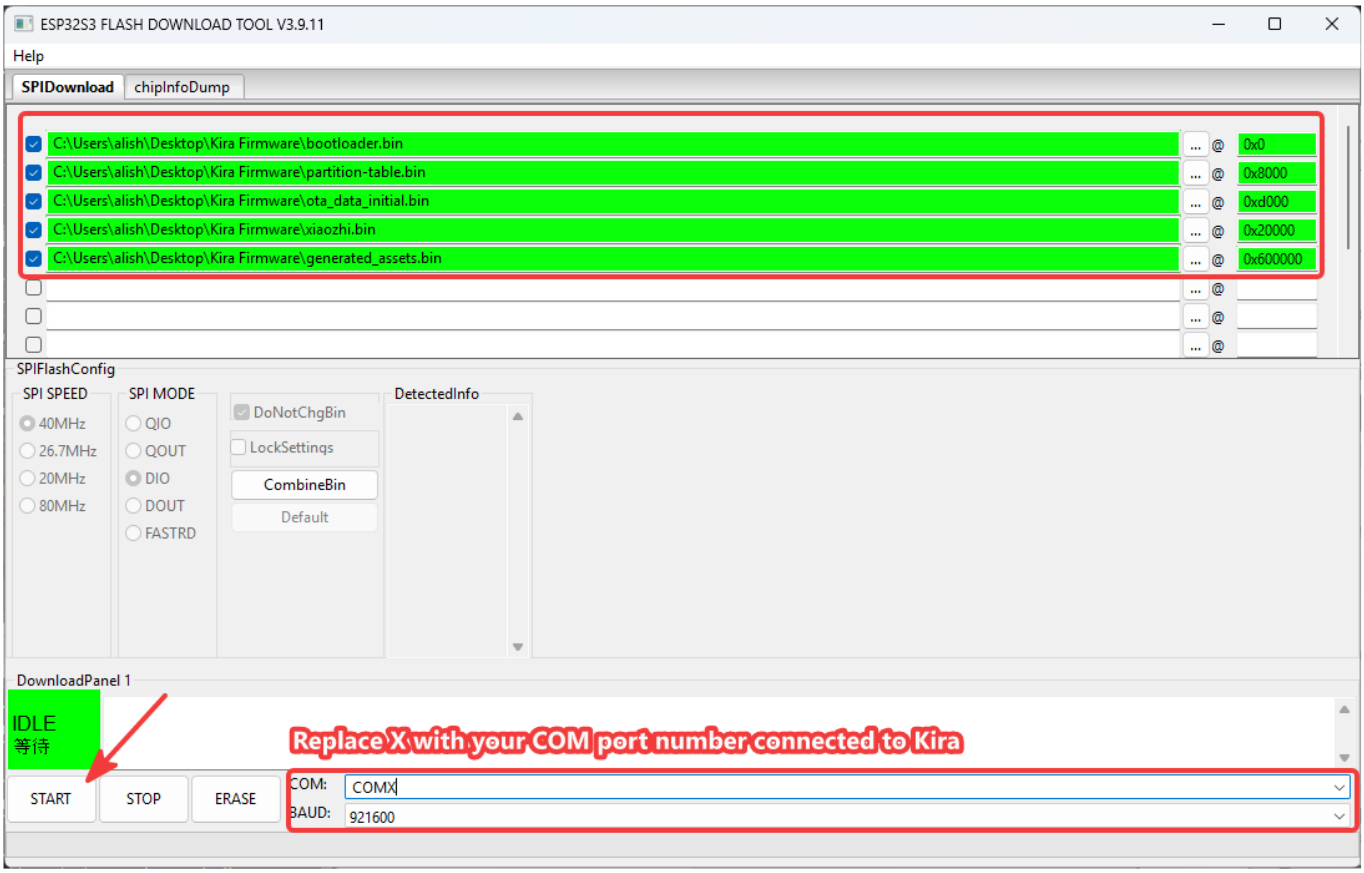
SAFETY NOTE

Power Kira only through a 5V USB port on your computer. Do not connect Kira directly to a wall outlet or mains power, as higher voltage can permanently damage the electronics.

6. The window will pop-up. Select the following options and hit ok.



7. You will see the following window open. Upload each bin file from the "Kira Firmware" folder and fill out the addresses (0x0, 0x8000 etc), COM port and BAUD as shown below. Then click START and wait.



8. Unplug Kira and plug again. You should see Kira's face and a sound.
9. Use your phone to connect to Xiaozhi Wi-Fi network.
10. The window should pop-up automatically. Enter your Wi-Fi network name into SSID and your network's password to Password. This will connect Kira to your Wi-Fi.
11. Go to xiaozhi.me and log-in using your gmail.
12. Click Add Device.
13. Kira will tell you the activation code. Enter the code and click Confirm.
14. Select Open Source, then click Start Uisng.
15. Configure the device settings as needed. You can setupp your preffered language.
16. Click Save.
17. Unplug Kira and plug it back again. It will reset.
18. Enjoy!

SAFETY NOTE

Power Kira only through a 5V USB port on your computer. Do not connect Kira directly to a wall outlet or mains power, as higher voltage can permanently damage the electronics.